Is Good Food Really Enough to Get 5 Stars on Google Maps?

What Drives Customer Satisfaction in Parisian Restaurants

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Abstract

This paper examines the key drivers of customer satisfaction in the Parisian restaurant industry by analyzing Google Maps reviews. Using a large language model to extract granular scores on six dimensions (food quality, value for money, service, wait time, atmosphere and cleanliness) from 5,000 reviews, we estimate their respective association with the star rating and with the overall sentiment of each review. Food quality is by far the most important factor ($\beta = 0.54$ for the rating), followed by service quality ($\beta = 0.26$). Value for money and atmosphere also play significant roles, while wait time and cleanliness have a smaller, though still positive, effect. The restaurant's popularity has no detectable influence. These findings suggest that while a holistic experience matters, culinary excellence and attentive service remain the primary pillars of restaurant success in competitive urban markets.

1 Introduction

Reputation on Google Maps has become a major concern for restaurant owners since the platform became a standard way of finding places to eat (Li and Hecht, 2020). More generally, in the digital economy, online reviews shape an establishment's reputation and have a measurable impact on revenue (Luca, 2016). Beyond the digital context, understanding customer reviews also requires the broader theory of consumer perception, which holds that the perceived value of an experience is a trade-off between price, quality and personal benefit (Zeithaml, 1988).

Google reviews are a particularly interesting case because ratings form the digital identity of an establishment. While exceptional food is traditionally the main reason people go to a restaurant, a dining experience is shaped by many other factors: a friendly waiter, a pleasant atmosphere or the perceived value for money can strongly influence how a customer feels. This study therefore investigates how these elements relate to the score a customer decides to leave, and to the overall positive or negative sentiment of the review.

Which aspects of a dining experience are most strongly associated with a good rating and a positive sentiment on Google Maps?

Figure 1 summarizes the hypothesized relationships. Based on them, we formulate four predictions:

P1 Food quality is the strongest predictor of a high rating, significantly outweighing the other factors.

P2 People tend to give higher ratings to restaurants that are already popular and have a large number of reviews.

P3 Since human interactions still matter, service quality is a strong factor explaining a positive sentiment.

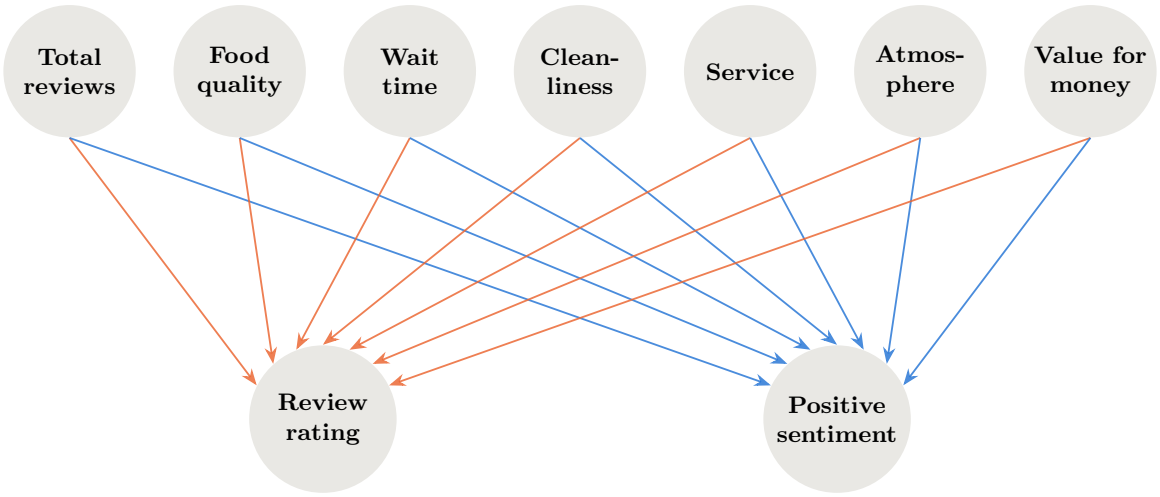


Figure 1: Hypothesized relationships (DAG). Orange arrows point to the review rating, blue arrows to the positive sentiment. Relationships between the criteria themselves are not shown, for readability.

P4 The restaurant setting, such as its atmosphere and cleanliness, is a strong predictor of a positive sentiment.

2 Data

The dataset was collected in March 2026. Using the Google Maps Places API, we first retrieved the identifier of each Parisian restaurant, along with its address, the number of user reviews it has received and the text of these reviews.

We then drew a representative sample of Parisian restaurants, selecting the same proportion of restaurants at each rating level as in the API results. The final dataset contains $N = 5{,}000$ reviews from about 3,800 restaurants (Figure 2). Each review was annotated by the Qwen2.5 language model, which assigned a score from -2 (very negative) to $+2$ (very positive), with 0 meaning neutral or not mentioned, for six criteria: **food quality**, **value for money**, **service**, **wait time**, **atmosphere** and **cleanliness**. The model also assigned a binary sentiment score (1 if the review is generally positive, 0 otherwise). To improve reliability, it was asked to justify its scores in one sentence and to score 0 any criterion not explicitly mentioned; the full prompt is given in Appendix A.

Each review is thus described by seven features: the six criteria above, plus the total number of reviews of the restaurant, used as a measure of its popularity.

3 Method

To test the predictions, we estimate two multiple linear regressions (OLS), explaining respectively the star rating given by the user and the overall sentiment of the review. All variables, outcomes included, are standardized beforehand, so each coefficient β is the change in the outcome, in standard deviations, associated with a one-standard-deviation increase in the predictor. This makes the effects of the different criteria directly comparable.

$$\begin{aligned}\text{Rating}_i = \beta_0 &+ \beta_1 \,\text{Food}_i + \beta_2 \,\text{Value}_i + \beta_3 \,\text{Service}_i + \beta_4 \,\text{Wait}_i + \beta_5 \,\text{Atmosphere}_i \\ &+ \beta_6 \,\text{Cleanliness}_i + \beta_7 \,\text{TotalReviews}_i + \varepsilon_i \end{aligned} \tag{1}$$

$$\begin{aligned}\text{Positive}_i = \beta_0 &+ \beta_1 \,\text{Food}_i + \beta_2 \,\text{Value}_i + \beta_3 \,\text{Service}_i + \beta_4 \,\text{Wait}_i + \beta_5 \,\text{Atmosphere}_i \\ &+ \beta_6 \,\text{Cleanliness}_i + \beta_7 \,\text{TotalReviews}_i + \varepsilon_i \end{aligned} \tag{2}$$

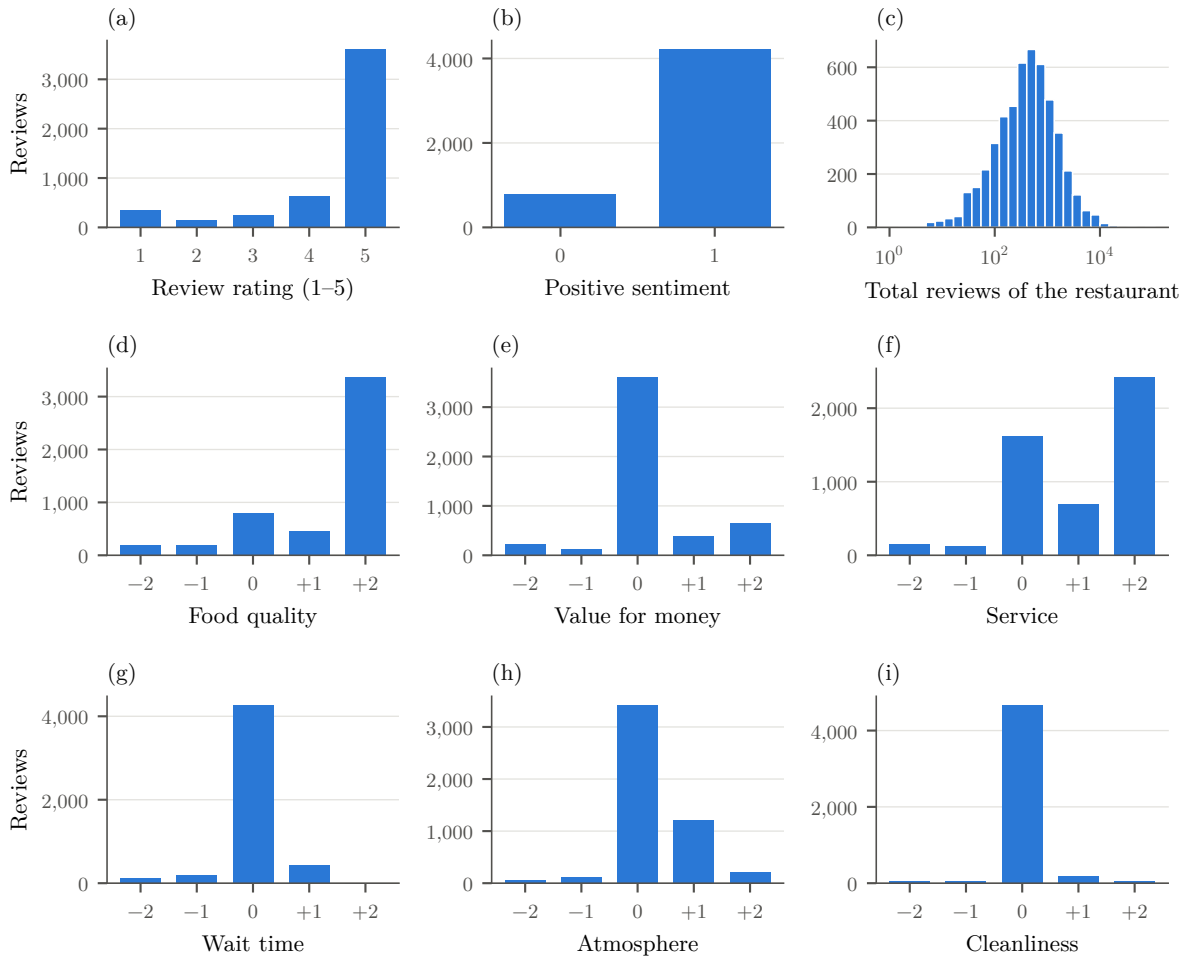


Figure 2: Univariate description of the dataset: (a) review rating, (b) positive sentiment, (c) total number of reviews of the restaurant (log scale), (d) food quality, (e) value for money, (f) service, (g) wait time, (h) atmosphere, (i) cleanliness. Criteria are scored from −2 to +2, with 0 when not mentioned.

4 Results

Table 1 and Figure 3 summarize both models.

4.1 Review rating

The first model explains a large share of the variance ($R^2 = 0.60$). As expected, food quality is the primary driver of the numerical rating ($\beta = 0.54$, $p < 0.001$; Figure 3a), which strongly supports **P1**: its coefficient is about twice that of any other criterion.

Beyond culinary attributes, the results confirm the importance of the human and physical dimensions of the dining experience. Service quality is the second most influential factor ($\beta = 0.26$, $p < 0.001$). Value for money ($\beta = 0.16$), atmosphere ($\beta = 0.10$), wait time ($\beta = 0.05$) and cleanliness ($\beta = 0.04$) are all positively and significantly associated with the star rating ($p < 0.001$).

P2, the "popularity effect", is not supported: the total number of reviews has no significant effect on individual ratings ($p = 0.227$). This suggests that customers tend to evaluate their experience on its own terms, regardless of the restaurant's existing reputation.

	(1) Review rating		(2) Positive sentiment	
Predictor	β	p	β	p
Food quality	0.538	< 0.001	0.566	< 0.001
Service	0.261	< 0.001	0.231	< 0.001
Value for money	0.156	< 0.001	0.127	< 0.001
Atmosphere	0.097	< 0.001	0.093	< 0.001
Wait time	0.047	< 0.001	0.058	< 0.001
Cleanliness	0.041	< 0.001	0.007	0.442
Total reviews	0.011	0.227	-0.010	0.287
R^2	0.60		0.58	

Table 1: Standardized coefficients of the two multiple regressions ($N = 5{,}000$ reviews, OLS).

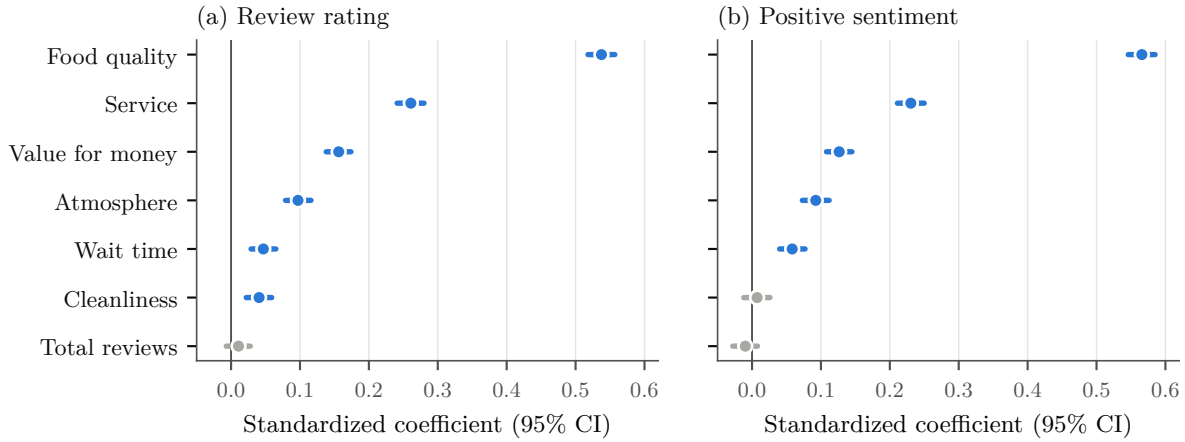


Figure 3: Standardized coefficients with 95% confidence intervals: (a) review rating, (b) positive sentiment. Grey marks a non-significant effect.

4.2 Positive sentiment

The second model ($R^2 = 0.58$) yields structurally similar results. Food quality ($\beta = 0.57$, $p < 0.001$; Figure 3b) and service ($\beta = 0.23$, $p < 0.001$) remain by far the dominant variables, which supports **P3**: service quality has a substantial association with a positive review. Value for money ($\beta = 0.13$), atmosphere ($\beta = 0.09$) and wait time ($\beta = 0.06$) remain significant ($p < 0.001$), while the restaurant's popularity again has no effect ($p = 0.287$).

The main difference between the two models concerns cleanliness. While it slightly influences the numerical rating, it is not significantly associated with an overall positive sentiment ($p = 0.442$). **P4** is therefore only partly supported: the physical setting matters, mostly through atmosphere, but cleanliness alone is not enough to secure a positive review. This may partly reflect how rarely cleanliness is mentioned at all (Figure 2i).

5 Conclusion

Customer satisfaction on Google Maps follows a clear hierarchy: food quality and service substantially outweigh value for money, atmosphere, wait time and cleanliness. While a holistic approach to service and setting is valuable, excelling in food quality and in the human side of service remains the most direct path to high ratings. Moreover, the absence of any association between a restaurant's review volume and its individual scores suggests that users judge their own experience rather than follow the crowd.

These results come with limitations. The scores are produced by a language model, whose

annotations may contain errors; most criteria are absent from most reviews, which limits the precision of their estimates; some restaurants contribute several reviews, so observations are not fully independent; and the regressions describe associations, not causal effects.

They also open new questions about the socio-geographical context. A promising next step would be to test whether these satisfaction drivers differ between tourist-heavy and residential districts of Paris. Exploring how location and local competition moderate these relationships would give a more nuanced understanding of urban consumer behavior in the digital age.

References

Li, Z., & Hecht, B. (2020). 3 stars on Yelp, 4 stars on Google Maps: A cross-platform examination of restaurant ratings. *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW2), 1–24.

Luca, M. (2016). Reviews, reputation, and revenue: The case of Yelp.com. *Harvard Business School Working Paper*, No. 12-016.

Zeithaml, V. A. (1988). Consumer perceptions of price, quality, and value: A means-end model and synthesis of evidence. *Journal of Marketing*, 52(3), 2–22.

A Annotation prompt

System prompt sent to the language model for review annotation (written in French).

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Rôle : Tu es un expert en linguistique et un moteur d'extraction de données pour une étude
sociologique sur la restauration. Ta tâche est de convertir des avis textuels en scores numériques
stricts.

RÈGLE D'OR ABSOLUE (ZÉRO DÉDUCTION) :
Si un critère n'est pas abordé explicitement et sans ambiguïté dans le texte, sa valeur est
OBLIGATOIREMENT 0. Ne fais aucune déduction logique (ex: une bonne nourriture ne signifie pas que
c'est propre).

Format de sortie : Uniquement un objet JSON valide. Aucune phrase d'introduction, aucune conclusion,
pas de balises markdown hors du JSON.

--- INSTRUCTIONS ET ÉCHELLES D'ANNOTATION ---
1. sentiment_positif (Impression globale)
 0 : Avis négatif, mitigé, déçu, ou purement factuel/neutre.
 1 : Avis clairement positif, enthousiaste, ou recommandation explicite.

2. score_nourriture (Qualité du produit)
-2 : Très mauvais (infect, cru, périmé, malade après repas, immangeable).
-1 : Décevant (fade, industriel, trop cuit, moyen, "sans plus").
 0 : Non mentionné.
 1 : Bon (savoureux, frais, fait maison, bien cuisiné).
 2 : Exceptionnel (délicieux, mémorable, "le meilleur", perfection gustative).

3. score_prix_valeur (Aspect financier)
-2 : Arnaque (vol, prix exorbitant, portions ridicules pour le prix).
-1 : Cher (abusif sur certains produits, rapport qualité/prix décevant).
 0 : Non mentionné (ou mention factuelle d'un prix standard sans jugement).
 1 : Correct (abordable, honnête, bon rapport qualité/prix).
 2 : Excellent (très bon marché, très copieux pour le prix, imbattable).

4. score_service (Rapport Humain)
Attention : Ne pas confondre avec le temps d'attente.
-2 : Odieux (agressif, insultant, raciste, ignore totalement le client).
-1 : Mauvais (froid, désagréable, amateur, impoli, serveur perdu).
 0 : Non mentionné (ou service standard syndical).
 1 : Bon (sympa, souriant, accueillant, pro, poli).
 2 : Parfait (aux petits soins, attentionné, chaleureux, conseil personnalisé).

5. score_attente (Gestion du Temps)
-2 : Interminable (plus d'1h, départ sans manger, scandaleusement long).
-1 : Lent (il a fallu réclamer, service désorganisé, longuet).
 0 : Non mentionné (rythme normal qui ne choque pas le client).
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 1 : Rapide (fluide, efficace, peu d'attente).
 2 : Éclair (servi immédiatement, bluffant de rapidité).

6. score_cadre_ambiance (Atmosphère & Lieu)
-2 : Insupportable (assourdissant, moche, sombre, tables collées, froid).
-1 : Pas terrible (déco vieillotte, musique forte, manque d'espace).
 0 : Non mentionné.
 1 : Agréable (joli, calme, bonne ambiance, charmant).
 2 : Magnifique (cadre idyllique, vue superbe, décoration incroyable).

7. score_proprete (Hygiène)
-2 : Danger (cafards, souris, toilettes immondes, vaisselle très sale).
-1 : Douteux (poussière, tables collantes, verres mal lavés, négligé).
 0 : Non mentionné (propre par défaut).
 1 : Propre (rien à signaler, net, bien tenu).
 2 : Impeccable (étincelant, propreté clinique, irréprochable).

8. justification
Rédige une seule phrase synthétique expliquant tes choix en citant les mots-clés de l'avis.

--- MODÈLE DE SORTIE JSON ATTENDU ---
{
  "sentiment_positif": 1,
  "score_nourriture": 1,
  "score_prix_valeur": 0,
  "score_service": 2,
  "score_attente": -1,
  "score_cadre_ambiance": 0,
  "score_proprete": 0,
  "justification": "Le client a aimé le plat (bon) et adoré le serveur (parfait), mais s'est plaint de
  la lenteur (-1). Le reste n'est pas mentionné (0)."
}
```